Nuclear Source and Reciprocal Alpha Receiving

Distribution-dependent fusion production and designed energy destinations

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Abstract

We construct direct nuclear-source and receiving calculations from the participating distributions. Exact shifted-ball geometry reduces the D–T reaction integral to one speed quadrature. Co-located material overlap makes the source evolve with the fuel. Full-Lorentz slowing assigns alpha energy and momentum to receiving species. Spatially shaped catalytic engagement supplies a distinct, dynamically tested wave channel. Together these operations make velocity distribution, spatial overlap, resonance and receiving interaction explicit variables of fusion research.

1. Distribution, transport and receiver define calculable design choices

Operational freedom. Nuclear production and its destinations can be calculated through distinct physical interactions, then connected through their complementary accounts. Distribution shape and drift enter the source; characteristic transport enters delivery; phase and resonance enter coherent exchange. Each calculation below identifies the parameters a researcher can vary and the receiver that makes the operation quantitative.

Recognitions R04-R06, R12-R13 and R21 share a dependency: a physical handoff must retain the donor, the interaction and the receiver. *From Closure to Composition* §3.1 requires the retained signature to carry the next admitted operation, while *From Participation to Quantitative Closure* requires complementary accounts. Here the actual distribution determines the reaction measure, the orbit determines transport, and the receiving species or field determines the usable form of deposited energy.

2. Exact shifted-ball reduction

At a fixed spatial point, the D and T distributions are uniform balls with radii a_v, b_v, center separation $d = |u_D - u_T|$, and volumes V_a, V_b. Let $p = a_v + b_v$ and $c_v = |a_v - b_v|$. Their centered difference density is the intersection volume divided by $V_a V_b$:

$$V_\cap(s) = \begin{cases} 4\pi \min(a_v, b_v)^3/3, & s \le c_v, \\ \pi(p-s)^2(s^2 + 2ps - 3c_v^2)/(12s), & c_v < s < p, \\ 0, & s \ge p. \end{cases} \tag{B1}$$

Angular integration gives the relative-speed probability density

$$P_d(g) = \frac{2\pi g}{dV_aV_b}\int_{|g-d|}^{g+d} sV_\cap(s)\,ds, \quad d>0; \qquad P_0(g) = \frac{4\pi g^2 V_\cap(g)}{V_aV_b}. \tag{B2}$$

The inner integral is piecewise polynomial. Its exact evaluation leaves a single numerical cross-section quadrature:

$$\mathcal{R}_{DT}(x) = n_D n_T \int_0^{a_v+b_v+d} P_d(g)\,\sigma_{DT}(\mu g^2/2) g\,dg, \qquad \mu = \frac{m_D m_T}{m_D+m_T}. \tag{B3}$$

The nonzero-drift support includes d. The zero-drift limit and equal-radius case are explicitly tested. Normalization and the second moment

$$\int P_d(g)dg = 1, \qquad \int g^2 P_d(g)dg = \tfrac{3}{5}(a_v^2+b_v^2)+d^2 \tag{B4}$$

provide independent geometric checks. The original implementation also compares against a 600,000-sample six-dimensional Monte Carlo calculation and a separate Maxwellian convolution. The latter does not make the waterbag Maxwellian.

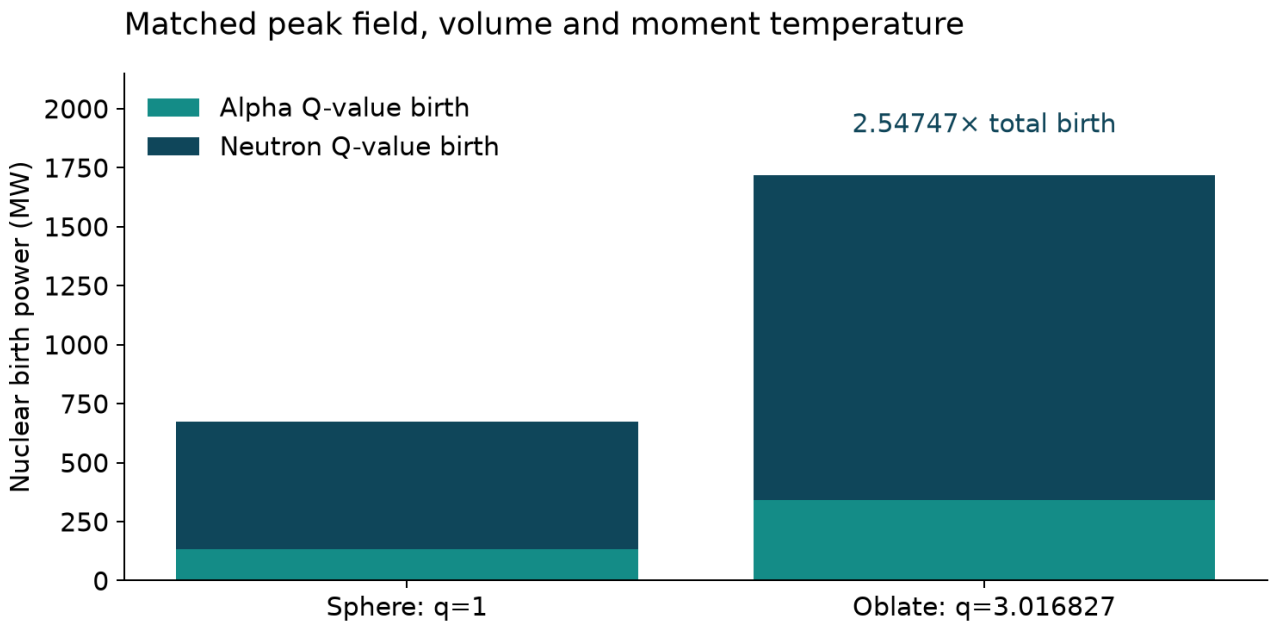
The supplied Bosch-Hale cross-section fit is an empirical nuclear input. The implementation specifies center-of-mass keV, millibarn-to-square-metre conversion and its accepted energy domain. Exactness refers to B1-B2 and their reduction, not to an exact fundamental nuclear cross section. An arbitrary anisotropic ellipsoid pair does not generally retain this one-dimensional formula; the later overlap constructor performs its own positive velocity quadrature.

3. A matched geometric birth comparison

The degree-nine specimen fixes peak field to 5 T, volume-equivalent radius to 1 m, volume to $4\pi/3$ cubic metres and peak electron moment temperature to 16.71 keV. The retained oblate shape is $q = 3.016827$, inherited as a test member from the earlier fluid search.

Account	Sphere	Oblate
Nuclear Q-value birth, MW	673.912	1716.768
Alpha birth, MW	134.017	341.403
Resistive current moment, MW	0.300910	0.487322
Mean electron density, $10^{21}\,\mathrm{m}^{-3}$	0.710131	1.18002

The ratio is **2.54746728** [C05-C07]. The differing inventories and field energies are essential comparison conditions. This is a selected nuclear birth functional, not equal-total-resource plant performance. Degree and field reconstruction errors are separate from the very small final speed-quadrature error.



Nuclear birth powers on the matched field-volume-temperature comparison.

4. Co-located source on moving material

For a positive material map $x_s(X,t)$ with deformation gradient F_s, conservation gives

$$n_s(x,t) = \frac{n_{s0}(X_s)}{\det F_s(X_s,t)}. \tag{B5}$$

The actual spatial overlap of D and T charts and their local velocity measures determines reaction events. It replaces a width-dependent localization kernel. Each event removes one D and one T, creates an alpha and a neutron, and retains momentum and Q-value accounting. The source constructor compares its undeformed limit with B3; deformed anisotropic cases retain their own quadrature and convergence requirements.

This operation is especially important in a nonlinear return model. Resetting density to its old value would be an undeclared material source. Real fuel admission requires a specified incoming phase-space measure, charge, energy and momentum. The capstone retains the implemented reaction debits without claiming an unconstructed fuel-processing plant.

5. Physical slowing and its receivers

The isotropic-waterbag collision construction evaluates Rosenbluth potentials from the actual compact velocity population. Its drift and diffusion then enter full Lorentz characteristics. The anisotropic extension uses the analytic isotropic result as a control variate and retains explicit species receiving.

For each collision increment, the complementary receiving material takes the lost alpha momentum and energy. Electromagnetic work is recorded separately. In the full-rate 16-history specimen, all sampled 3.5 MeV histories reach the 200 keV cutoff after approximately 35.7 ms and yield about 321.18 MW scaled species deposition [C08]. The fraction near 94.08% is consistent with the large birth-to-cutoff energy interval, with the actual field and collision accounts retained. It is not an independent claim that an arbitrary alpha population has that confinement fraction. The 64-history result in the later carrier is a separate ensemble.

The distinction between birth and receiving matters quantitatively. A large alpha birth-to-resistive-moment ratio does not derive a coherent current-renewal channel. Nor does species heating establish net useful electrical work. Those operations require their actual conversion and support relations.

6. Catalytic engagement with a derived spatial map

The integrated slab construction uses a potential

$$\Phi = \frac{E_0}{|k_y|} h(x)\cos(k_y y + k_\parallel z - \omega t), \qquad h(x) = \sum_\ell b_\ell e^{iq_\ell x}. \tag{B6}$$

Its gyro-orbit harmonic is calculated along the entire orbit:

$$C_n(X,\rho) = \frac{1}{2\pi}\int_0^{2\pi} h(X + \rho\cos\varphi) e^{ik_y\rho\sin\varphi - in\varphi}\, d\varphi. \tag{B7}$$

A Gaussian potential notch suppresses core coupling but does not delete reverse transitions. The same sidebands determine resonant alpha diffusion and receiver absorption. For a canonical transition edge,

$$J_{hl} = E_0^2 G_{hl}\,(N_h/g_h - N_l/g_l), \quad g(E) \propto \sqrt{E}, \quad \Delta x = \frac{k_y}{q_\alpha B\omega}\Delta E. \tag{B8}$$

Opposite population updates conserve count internally; the energy and canonical transverse momentum have complementary field or boundary recipients. Thermalization retains spatial position. The selected equal-D/T fuel return is a supplied material closure; quasineutrality alone does not uniquely derive a fuel pinch.

The catalytic wave has its own energy account and receiver. Bulk ion auxiliary input is a separate port, and electrons float thermally without an active temperature-restoring heater. The 550 keV notched case gives a bounded 30 s seed-free channel endpoint [C01]. The 560 keV higher-gain equilibrium fails the tested startup admission. Neither demonstrates the hypothesized decrease in fast-alpha resident inventory. These retained counterexamples make the design operation usable: spatial shaping can remove desirable receiver damping as well as parasitic donor reheating.

7. Source and receiving as an executable design space

What this opens. Velocity-space shape and drift can enter reaction calculation directly. Material deformation can change the source through actual overlap. Species receiving can be traced along characteristic histories, and a separately constituted wave channel can be designed through spatial and resonant coupling. These are four concrete operations available for comparison and composition.

The exact reaction measure, co-located source and collision receiving are usable numerical constructors. The slab potential is a conditional interaction family with a supplied dielectric energy normalization; it is not a solved global launcher and mode. A source-to-orbit-to-wave composition must therefore supply that map, retain pitch and parallel momentum where relevant, and recompute receiving and loss under the same geometry.

For reproduction, use demonstration sets `reaction`, `deposition`, `channel` and `source`. The authoritative JSON pointers for the selected numbers are in the claim register; the full recipes remain in the archived READMEs. Each accounts for its own physical carrier. A researcher can now calculate how a changed distribution, spatial engagement or receiver alters production and energy delivery through an explicit interaction.

Source navigation: Bibliography · claim-to-receipt register · demonstration recipes.